

LUHANSK INTL, Ukraine

WMO: 330001

Lat: 48.417N

Lon: 39.374E

Elev: 194

StdP: 99.02

Time Zone: 2.00 (EUE)

Period: 04-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-19.0	-24.8	0.4	-21.8	-21.2	0.6	-18.5	12.8	-4.3	11.7	-3.3	3.3	60	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	35.0	19.1	33.0	18.9	31.0	18.5	21.2	28.5	20.6	28.1	19.9	27.5	5.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	23.6	18.1	13.4	22.9	17.2	12.6	22.4	62.4	29.2	60.1	27.9	57.7	27.4	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.0	9.0	DB	-23.4	37.2	4.2	1.7	-26.4	38.5	-28.9	39.5	-31.2	40.4	-34.2	41.7	(4)
(5)			WB	-23.6	23.1	4.1	1.1	-26.6	23.9	-29.0	24.5	-31.4	25.1	-34.4	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-5.3	-5.1	1.3	9.9	17.2	20.6	22.8	22.9	16.2	9.5	3.3	-1.7	(6)	
	DBStd	11.26	7.19	6.83	4.62	4.21	4.35	3.45	3.36	3.98	3.71	4.40	4.64	5.41	(7)	
	HDD10.0	1861	475	423	271	52	4	0	0	0	2	64	206	364	(8)	
	HDD18.3	3703	733	657	528	254	75	16	4	7	82	275	451	622	(9)	
	CDD10.0	1632	0	0	1	49	226	318	396	399	189	48	5	0	(10)	
	CDD18.3	433	0	0	0	1	38	84	142	148	19	1	0	0	(11)	
	CDH23.3	4733	0	0	0	21	429	871	1489	1713	207	4	0	0	(12)	
	CDH26.7	1949	0	0	0	2	128	312	641	817	49	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.9	5.0	5.2	4.4	3.7	3.5	3.2	3.4	3.7	4.0	4.3	5.0	(14)	
(15) Precipitation	PrecAvg	533	41	37	38	34	46	62	61	34	47	40	41	51	(15)	
	PrecMax	739	86	95	74	68	105	131	153	79	126	92	77	115	(16)	
	PrecMin	364	16	12	17	6	4	16	5	4	3	16	3	21	(17)	
	PrecStd	108	20	19	15	16	23	32	39	18	33	20	19	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	6.9	16.2	26.1	32.9	34.1	36.9	38.8	30.9	23.0	17.1	11.0	(19)	
		MCWB	7.0	4.4	9.4	14.3	18.6	19.3	18.8	19.0	16.6	13.9	12.1	10.5	(20)	
	2%	DB	5.9	5.0	12.9	22.2	29.9	32.2	34.2	36.1	27.9	20.2	13.9	8.1	(21)	
		MCWB	5.1	3.9	7.5	13.0	17.6	19.1	19.3	18.9	17.2	13.9	10.5	7.3	(22)	
	5%	DB	4.2	3.8	10.0	19.9	27.9	30.1	32.2	33.9	25.8	18.1	11.2	6.2	(23)	
		MCWB	3.5	3.1	6.5	11.9	17.0	18.4	19.4	18.4	16.1	13.3	9.1	5.7	(24)	
	10%	DB	2.8	2.8	7.9	17.2	25.8	28.1	30.2	31.1	23.8	16.0	9.8	4.9	(25)	
		MCWB	2.4	2.2	5.6	10.6	16.0	18.1	19.0	18.2	15.1	13.1	8.3	4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	5.6	10.5	15.6	20.5	21.9	22.9	21.6	20.1	17.1	12.6	10.6	(27)	
		MCDB	7.8	6.0	14.6	24.0	29.2	28.2	31.7	30.6	25.7	21.0	15.7	11.2	(28)	
	2%	WB	5.1	4.0	8.6	14.0	18.9	20.8	21.5	20.7	18.5	15.4	10.9	7.2	(29)	
		MCDB	5.4	4.5	11.5	21.0	26.7	28.0	28.3	29.3	24.2	18.5	13.4	7.7	(30)	
	5%	WB	3.5	2.9	7.0	12.7	18.0	20.0	20.9	20.0	17.3	14.3	9.7	5.8	(31)	
		MCDB	4.0	3.4	9.0	17.8	25.0	27.6	27.8	28.7	22.8	16.8	11.2	6.2	(32)	
	10%	WB	2.1	1.9	5.5	11.5	17.0	19.3	20.1	19.3	16.2	13.2	8.2	4.3	(33)	
		MCDB	2.5	2.3	7.5	16.0	23.8	26.2	27.1	28.1	21.2	15.4	9.3	4.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.0	7.8	10.7	11.9	11.9	12.1	13.1	11.6	8.5	6.2	4.5	(35)	
		MCDBR	5.2	6.0	11.5	14.2	14.7	14.5	14.5	15.9	14.5	11.1	9.0	5.3	(36)	
	5% WB	MCWBR	4.7	5.1	7.4	7.3	6.2	5.5	5.2	5.4	6.4	6.3	6.2	4.7	(37)	
		MCWBR	5.1	4.9	10.0	11.8	12.5	12.2	11.9	12.9	11.9	9.6	7.5	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.358	0.390	0.448	0.413	0.411	0.435	0.436	0.406	0.374	0.347	0.331	(40)	
		taud	2.194	2.115	2.158	2.071	2.262	2.298	2.241	2.220	2.269	2.322	2.305	2.231	(41)	
	Edn at Noon		706	767	811	799	850	853	824	803	789	752	685	649	(42)	
		Edh at Noon	81	111	125	151	130	126	132	129	112	90	74	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.99	1.93	2.97	4.34	5.82	6.29	6.14	5.61	3.91	2.23	1.04	0.64	(44)	
	RadStd		0.22	0.40	0.47	0.47	0.52	0.53	0.39	0.39	0.38	0.34	0.20	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (7 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-173	+117	+74	(47)	

Nomenclature: See separate page